Technical Requirements Document

Workforce & Payroll Management Platform

Project: Team 97 | IITM Software Engineering Course **Version:** 1.0 | Date: 2026-03-06

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1. System Overview

A web-based workforce and payroll management system for a dark store / warehouse with 30–35 employees operating across 3 daily shifts. The system replaces manual registers and Excel sheets with a centralized platform for attendance, leave, incentive, advance, and salary management.

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2. User Roles

Role	Permissions
`owner`	Full access — manage employees, record attendance, approve advances, compute salaries
`supervisor`	Record attendance, log incentives, view employee data
`employee`	View own attendance, payslip, leave balance, advance status

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3. Functional Requirements

FR-1: Authentication

- FR-1.1: All users must log in with username and password.
- FR-1.2: Sessions are role-based; unauthorized routes return HTTP 403.
- FR-1.3: Owner account is seeded at setup; employees are created by the owner.

FR-2: Employee Management

- FR-2.1: Owner can add, edit, and deactivate employee records.
- FR-2.2: Each employee record stores: name, phone, role, shift assignment, base daily wage, and join date.

- FR-2.3: Employees are assigned to one of three shifts: Shift 1, Shift 2, Shift 3.

FR-3: Attendance

- FR-3.1: Attendance is recorded once per employee per day by the owner or supervisor.
- FR-3.2: Attendance status must be one of: `present`, `absent`, `half_day`, `on_leave`.
- FR-3.3: Hours worked must be recorded (standard shift = 8.5 hours net).
- FR-3.4: If hours worked > 8.5, the excess is classified as overtime.
- FR-3.5: Duplicate attendance records for the same employee on the same date are rejected.

FR-4: Leave Management

- FR-4.1: Each employee is entitled to 4 paid leaves per calendar month.
- FR-4.2: Attendance status `on_leave` counts against the paid leave quota.
- FR-4.3: Leaves beyond 4 per month are classified as unpaid and trigger a salary deduction equal to one day's base wage per excess leave.
- FR-4.4: Owner can view leave counts per employee per month.

FR-5: Incentives

- FR-5.1: An incentive entry is recorded per employee per day (one entry per attendance day).
- FR-5.2: Incentive category must be one of: `incentive_1`, `no_incentive`, `incentive_2`.
- FR-5.3: Incentive monetary values are configurable by the owner (stored as system settings).
- FR-5.4: Monthly incentive total = sum of all daily incentive amounts for that month.

FR-6: Salary Advances

- FR-6.1: Employees can request a salary advance; request is logged with the request date.
- FR-6.2: Eligibility rule: eligible advance amount = (request_day - 7) * daily_wage.
  - Example: Request on the 10th → eligible = 3 days' salary.
  - Requests on day 1–7 of the month are ineligible (result: 0).
- FR-6.3: Owner approves or rejects advance requests.
- FR-6.4: Approved advance amount is deducted during month-end salary computation.
- FR-6.5: Only one active (unapproved or approved-but-not-yet-deducted) advance per employee per month is permitted.

FR-7: Salary Computation

- FR-7.1: Owner triggers salary computation for a given month via a single action.
- FR-7.2: Salary formula per employee:

```
gross_salary = days_present * daily_wage
             + (overtime_hours * overtime_rate)
             + monthly_incentive_total

deductions   = unpaid_leave_days * daily_wage
             + approved_advance_amount

net_salary   = gross_salary - deductions
```

- FR-7.3: `days_present` counts attendance records with status `present` or `half_day` (half_day = 0.5 days).
- FR-7.4: `overtime_hours` = sum of (hours_worked - 8.5) for all days where hours_worked > 8.5.
- FR-7.5: Overtime rate is configurable (stored as system setting, e.g., 1.5x hourly wage).
- FR-7.6: Computed salary records are stored and cannot be overwritten once finalized; owner must explicitly re-compute if corrections are needed.
- FR-7.7: Each finalized salary record constitutes a payslip viewable by the employee.

FR-8: Reporting / Views

- FR-8.1: Owner can view a monthly attendance summary table (all employees x all days).
- FR-8.2: Owner can view a payroll summary table for any finalized month.
- FR-8.3: Employee can view their own monthly payslip (breakdown of all components).

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4. Business Rules Summary

Rule	Detail
Shift duration	9 hours gross, 8.5 hours net
Paid leaves	4 per employee per month
Excess leave deduction	1 day's base wage per extra leave
Overtime threshold	> 8.5 hours worked in a day
Advance eligibility	(request_day_of_month - 7) * daily_wage

Incentive categories	`incentive_1`, `no_incentive`, `incentive_2`
Half-day attendance	Counts as 0.5 days for salary

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5. Non-Functional Requirements

- NFR-1: The system must support up to 35 concurrent employee records.
- NFR-2: All salary computations must be deterministic and reproducible given the same input data.
- NFR-3: Passwords must be stored as hashed values (bcrypt or werkzeug).
- NFR-4: The API must return JSON responses for all endpoints.
- NFR-5: The application must run locally (no cloud deployment required for this phase).

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6. Out of Scope

- Biometric or hardware-based attendance capture
- Mobile application
- Email / SMS notifications
- Tax (TDS) or statutory deduction computation (PF, ESI)
- Multi-location / multi-store support